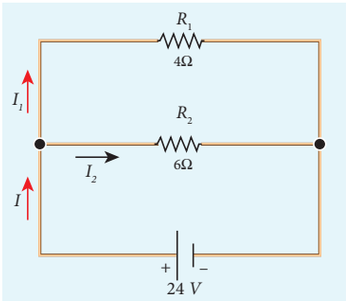
EXAMPLE 2.9

Calculate the equivalent resistance in the following circuit and also find the current I, I_1 and I_2 in the given circuit.



Solution

Since the resistances are connected in parallel, therefore, the equivalent resistance in the circuit is

$$\frac{1}{R_P} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{4} + \frac{1}{6}$$

$$\frac{1}{R_P} = \frac{5}{12}\Omega \quad \text{or } R_P = \frac{12}{5}\Omega$$

The resistors are connected in parallel, the potential (voltage) across each resistor is the same.

$$I_1 = \frac{V}{R_1} = \frac{24V}{6\Omega} = 6A$$

$$I_2 = \frac{V}{R_2} = \frac{24}{6} = 4A$$

The current I is the total of the currents in the two branches. Then,

$I = I_1 + I_2 = 6\text{ A} + 4\text{ A} = 10\text{ A}$

EXAMPLE 2.10

When two resistances connected in series and parallel their equivalent resistances are 15 Ω and $\frac{56}{15}\Omega$ respectively. Find the individual resistances.

Solution

$$R_s = R_1 + R_2 = 15\ \Omega \qquad (1)$$

$$R_P = \frac{R_1 R_2}{R_1 + R_2} = \frac{56}{15}\Omega \qquad (2)$$

From equation (1) substituting for $R_1 + R_2$ in equation (2)

$$\frac{R_1 R_2}{15} = \frac{56}{15}\Omega$$

$$\therefore R_1 R_2 = 56$$

$$R_2 = \frac{56}{15}\Omega \qquad (3)$$

Substituting for R_2 in equation (1) from equation (3)

$$R_1 + \frac{56}{R_1} = 15$$

Then, $\frac{R_1^2 + 56}{R_1} = 15$

$$R_1^2 + 56 = 15\ R_1$$

$$R_1^2 - 15\ R_1 + 56 = 0$$

The above equation can be solved using factorisation.

$$R_1^2 - 8\ R_1 - 7\ R_1 + 56 = 0$$

$$R_1\ (R_1 - 8) - 7\ (R_1 - 8) = 0$$

$$(R_1 - 8)\ (R_1 - 7) = 0$$

If $(R_1 = 8\ \Omega)$

using in equation (1)

$$8 + R_2 = 15$$

$$R_2 = 15 - 8 = 7\ \Omega\ ,$$

$R_2 = 7\ \Omega$ i.e , (when $R_1 = 8\ \Omega$; $R_2 = 7\ \Omega$)

If $(R_1 = 7\ \Omega)$

Substituting in equation (1)

$$7 + R_2 = 15$$

$R_2 = 8\ \Omega$, i.e , (when $R_1 = 8\ \Omega$; $R_2 = 7\ \Omega$)